$$\text{Thus, } I = \frac{1}{2}\int \frac{1}{\sqrt{t^2-5^2}}\,dt$$

$$= \frac{1}{2}\log\left|t+\sqrt{t^2-5^2}\right|+c$$

$$I = \frac{1}{2}\log\left|2x+\sqrt{4x^2-25}\right|+c$$

Type II

Integrals of the form $\int \frac{dx}{ax^2+bx+c}$ and $\int \frac{dx}{\sqrt{ax^2+bx+c}}$

First we express ax^2+bx+c as sum or difference of two square terms that is, any one of the forms to Type I. The following rule is used to express the expression ax^2+bx+c as a sum or difference of two square terms.

(1) Make the coefficient of x^2 as unity.

(2) Completing the square by adding and subtracting the square of half of the coefficient of x.

$$\text{That is, } ax^2+bx+c = a\left[x^2+\frac{b}{a}x+\frac{c}{a}\right]$$

$$= a\left[\left(x+\frac{b}{2a}\right)^2+\frac{4ac-b^2}{4a^2}\right]$$

Examples 11.39

Evaluate the following integrals

(i) $\int \frac{1}{x^2-2x+5}\,dx$ (ii) $\int \frac{1}{\sqrt{x^2+12x+11}}\,dx$ (iii) $\int \frac{1}{\sqrt{12+4x-x^2}}\,dx$

Solution

(i) $$\int \frac{1}{x^2-2x+5}\,dx = \int \frac{1}{x^2-2(1)x+(1)^2+4}\,dx$$

$$= \int \frac{1}{(x-1)^2+2^2}\,dx$$

$$\int \frac{1}{x^2-2x+5}\,dx = \frac{1}{2}\tan^{-1}\left(\frac{x-1}{2}\right)+c$$

(ii) $$\int \frac{1}{\sqrt{x^2+12x+11}}\,dx = \int \frac{1}{\sqrt{(x+6)^2-25}}\,dx$$

$$= \int \frac{1}{\sqrt{(x+6)^2-5^2}}\,dx$$

$$= \log\left|x+6+\sqrt{(x+6)^2-5^2}\right|+c$$

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